PhyloTrace

PRODUCT DATASHEET

Genomes and epidemiology in one platform.

PhyloTrace is an open-source platform for bacterial pathogen surveillance and monitoring. It integrates genomic typing, resistance screening and epidemiological metadata — ongoing and retrospective, in public health, healthcare, food-associated and environmental settings — on your own hardware.

1. **Genomes and epidemiology together**
   Typing, resistance and your metadata analysed side by side.
2. **Ongoing and retrospective**
   Monitor as isolates arrive or analyse existing collections.
3. **Comparable without a central server**
   Allele fingerprints give partners identical results.
4. **Verifiable and under your control**
   Provenance for every result, GPLv3, runs locally.

Key features

cgMLST & MLST typing

More than 40 ready-to-use schemes cover the most relevant bacterial species, typed fast and automatically, with classical MLST alongside for comparability with published types — so comparisons start from clean data.

Resistance & virulence

NCBI AMRFinderPlus screens every assembly for resistance genes, virulence factors, species-specific point mutations and stress, biocide and metal tolerance — so resistance is part of every analysis.

Overviews & quality control

Overviews of provenance, assembly quality, completeness and the settings of every run, plus consistency and duplicate checks — so weak data never distorts clusters.

Custom metadata

Any variable — wards, sources, sites, susceptibility — checked on entry and mapped onto every view, so relatedness is always read in its epidemiological context.

Visual analysis

MSTs, trees, epidemic curves, maps, heatmaps, annotated with any variable — so you see when and where transmission happened.

Sharing & dashboard

Saved analyses and selective, field-by-field exchange between laboratories — so routine monitoring is repeatable and you decide what leaves your lab.

Decentralised by design

Local analysis
Data stays on your hardware, where the samples are.

Direct comparison
Partner databases merge on allele fingerprints.

Selective sharing
You choose which isolates and fields travel.

Independent
Keeps working when a partner or hub is down.

Technical specifications

Runs on	Linux; Windows support planned. Desktop workstation (tested: 8 cores, 24 GB RAM).
Interface	Local application in your web browser; all computation and data stay on your machine.
Installation	One installer for PhyloTrace and all analysis tools it uses.
Input	Bacterial genome assemblies (FASTA). Integrated genome assembly is planned.
Typing	cgMLST (42 species schemes), classical MLST
Screening	NCBI AMRFinderPlus: resistance genes, virulence, stress/biocide/metal tolerance, point mutations
Data	One portable database file per project
Allele identity	SHA-256 hash of each allele sequence; partner databases merge directly
Exchange & figures	Database file, Excel, CSV/TSV, FASTA · PDF, SVG, PNG
Licence	GNU GPL v3 · Zenodo DOI 10.5281/zenodo.10996423

Plans

Community · Free
Software, public updates, public documentation, community support on GitHub

Subscription · €149/month or €1,499/year[1]
Direct support (target first response 1 business day for critical issues, 2–5 business days otherwise), priority and earlier access to updates, installation, remote onboarding, professional documentation

Getting started

1 Download
Free from GitHub

2 Demo
For your use case

3 Pilot
6 months

4 Subscription
Routine use

Pilot: 6 months at **€99/month**[1], credited in full against the first year of the annual subscription.

Trust

Designed with data protection in mind: genomes and metadata stay on your hardware, no customer data at Liora, no telemetry · Open source under GPLv3 · Developed with HFU and MUG · Made in Germany

Request a demo
info@liora-bioinformatics.com
liora-bioinformatics.com/phylotrace

[1] All prices are net prices in euros, plus statutory VAT; subscription with a minimum term of 12 months.

For research use only. PhyloTrace is not a medical device and is not intended for diagnostic procedures, treatment decisions or the management of individual patients.
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